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Course 608

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Evaluating the Efficacy of dCas9-KRAB for Transient TTR Gene Silencing in Early Mammalian Embryos

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Introduction

Hereditary transthyretin amyloidosis (hATTR) is a dominantly inherited disease caused by pathogenic gene mutations in the transthyretin (TTR) gene [1]. TTR expresses a transporter gene in the liver that binds thyroid hormone and retinol-binding protein [2]. Defect-inducing point mutations destabilize the native TTR tetramer so it fragments into single monomers, which further misassemble into toxic amyloid fibrils [2] [3]. These insoluble fibrils accumulate in the nerve and myocardial tissue causing severe polyneuropathy and cardiomyopathy [1].
New RNA interference drugs and recently developed adult in vivo CRISPR-Cas9 nuclease (NTLA-2001) gene editing therapies are only able to intervene after tissue destruction occurs [4]. Nuclease-active CRISPR-based platforms depend on intentionally induced double-strand DNA breaks, which are unacceptable in the setting of early embryonic life where specific off-target cleavage is inevitable as is the risk of chromosomal mosaicism [4] [5].
Evaluating a non-cutting dCas9-KRAB approach at the single-cell stage is critical because it stops toxic protein accumulation long before organ damage can begin. Furthermore, proving that target genes can be silenced without breaking DNA provides a safer alternative to traditional CRISPR endonucleases, and builds a generalizable framework for treating various dominant inherited diseases.

Research Problem

Hereditary transthyretin amyloidosis (hATTR) is a progressive disease caused by autosomal dominant mutations in the *TTR* gene, which lead to misfolded transthyretin protein that breaks down and forms toxic protein clumps in nerve and heart tissue [10]. A recent systematic review estimated global prevalence at roughly 50,000 people, with wide regional variation and endemic

clusters in Portugal, Sweden, and Japan [1]. Current treatments, including RNA-silencing drugs and an *in vivo* CRISPR-Cas9 therapy (NTLA-2001) that permanently knocks out *TTR* in adult liver cells, reduce circulating TTR only after disease has already progressed into adulthood, and rely on permanently cutting DNA [2].

Because NTLA-2001 physically cuts the genome, its effects cannot be reversed when applied [3]. Applying DNA-cutting editing this early in life, at the embryonic stage, would carry immense risks of off-target double-strand breaks and mosaicism, where only some cells of the developing embryo carry the intended edit [4]. This raises the question of whether a non-cleavage form of gene control could someday allow *TTR* expression to be dialed down long before misfolded protein has had years to accumulate, without permanently altering the genome. Epigenetic repression using a disabled Cas9 that turns off genes without cutting DNA (dCas9-KRAB) offers exactly this kind of reversible alternative [5], [6], but whether it can silence *TTR* specifically, and whether that silencing is safe and stable when applied at the single-cell zygote stage, has not yet been tested.

Existing Literature

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The first human trial of in vivo CRISPR-Cas9 genome editing targeted *TTR* directly and reported an 87% mean reduction in serum TTR protein following a single dose, demonstrating that lowering TTR expression is a clinically meaningful therapeutic strategy [2]. Subsequent commentary on this trial noted that because the treatment permanently deletes genomic sequence, it cannot be titrated or reversed once given [3].

Independent of hATTR research, earlier CRISPR-Cas9 work in mammalian zygotes showed that nuclease-active editing can correct or disrupt target genes in early embryos [4]. But this frequently produces mosaic embryos in which not every cell carries the intended edit, alongside a risk of unintended off-target cuts [4]. To avoid these problems, dCas9 was developed as a catalytically dead version of Cas9 that binds a target DNA sequence without cutting it, functioning solely as a programmable delivery vessel [5]. Fusing dCas9 to the KRAB repressor domain was shown to enable sequence-specific transcriptional silencing in adult somatic cells by recruiting enzymes that deposit repressive H3K9me3 chromatin marks at the target location [6].

However, early mammalian embryos undergo extensive epigenetic reprogramming during the first cell divisions, and this reprogramming does not appear to erase all loci uniformly [7]. Some regions retain repressive or active marks depending on sequence context [7]. No published study has computationally assessed guide RNA specificity or chromatin-retention likelihood for a dCas9-KRAB approach targeting *TTR* specifically at the zygote stage. Structurally, TTR is a four-subunit carrier protein whose toxic accumulation depends on the dissociation of its native tetramer into single monomers, which subsequently misfold and form amyloid fibrils [10]. Publicly available data in the Protein Data Bank provides the necessary foundation to model how reducing *TTR* expression alters this breakdown pathway [11].

Research Question

Based on publicly available genomic, epigenomic, and structural datasets, how feasible is dCas9-KRAB-mediated transcriptional repression of the *TTR* gene at the zygote stage, in terms of (a) guide RNA specificity and predicted off-target binding, (b) the likelihood that KRAB-deposited repressive marks would persist through known patterns of early embryonic epigenetic reprogramming, and (c) the structural and functional consequences of reduced TTR protein expression?

Methodology

17 Candidate single guide RNAs targeting the *TTR* promoter region (100–200 bp upstream of the transcription start site) will be designed using reference mouse and human genome sequences available through Ensembl and NCBI. Candidates will be generated and scored using CRISPOR, a guide-design and off-target prediction tool validated against experimental off-target datasets [8], and cross-checked using CHOPCHOP, an independently developed web tool for CRISPR guide selection [9]. Each candidate will be ranked by predicted on-target specificity score and the number and severity of predicted genome-wide off-target binding sites.
Publicly deposited single-cell embryo epigenomic datasets covering histone modification states across mouse pre-implantation stages will be analyzed to determine whether the *TTR* promoter region and its surrounding chromatin are likely to retain or lose repressive marks during the developmental window relevant to this proposal, building on existing evidence that reprogramming does not erase repressive chromatin uniformly across the genome [7].

Publicly available TTR protein structural data from the Protein Data Bank [11] will be used to review how reduced TTR expression is expected to affect tetramer stability and amyloidogenic misfolding, drawing on established structural findings [10] rather than new wet-lab assays. Guide RNA candidates will be ranked and filtered by specificity score. Epigenomic retention likelihood will be summarized descriptively from the public datasets reviewed. Findings across all three phases will be synthesized into an overall feasibility assessment addressing the research question's three components. This study uses only publicly available, de-identified genomic, epigenomic, and structural datasets. No animal or human embryos are used or created at any stage.

Research Topic

This research applies a clinically validated therapeutic target, *TTR*, to a fundamentally different editing strategy at the earliest possible developmental stage. Where the existing human trial relies on permanent nuclease-based knockout in adult liver cells, this project computationally tests whether the same disease-relevant gene could plausibly be silenced far earlier in development, offering an evidence-based case for (or against) pursuing this reversible, non-destructive strategy experimentally in the future.

Project Practicalities

All data sources used (Ensembl, NCBI, GEO, PDB) are freely and publicly accessible, and all analysis tools, including CRISPOR [8] and CHOPCHOP [9], run through a web browser or on a personal computer, so no lab access, animal use, or specialized equipment is required. The project is expected to take four to six weeks. Roughly one to two weeks for guide RNA design and scoring, two weeks for epigenomic dataset analysis, and one to two weeks for structural literature synthesis and write-up.

Roadblocks and potential limitations

Guide RNA off-target prediction tools like CRISPOR and CHOPCHOP are built and validated primarily on data from somatic cell lines, not zygotes, so their specificity scores may not transfer cleanly to the very different chromatin state of a single-cell embryo [8] [9]. Second, public epigenomic datasets on mouse pre-implantation embryos are sparse at the single-locus resolution needed to say anything precise about the *TTR* promoter specifically; it is entirely possible the available data simply won't be granular enough to answer part (b) of the research question with confidence, in which case the honest outcome would be "inconclusive" rather than yes/no. Third, because no wet lab work is being done, this proposal cannot verify that a computationally specific guide RNA would actually behave as predicted once delivered to a real embryo, and I want to be upfront that a favorable computational result is a reason to pursue lab testing, not a substitute for it. Finally, my own inexperience with bioinformatics tooling means the first few weeks will likely involve a learning curve with genome browsers, which could compress the time available for the epigenomic analysis. I intend to spend the first week familiarizing myself with the tools to mitigate this risk.

Post-program plan

When completed, this analysis will be submitted to student research showcases. Depending on how the results turn out, I plan to look into submitting a complete version to science journals such as the Youth Research Journal (my top choice), or the Journal of Emerging Investigators. I also intend to use this project as the basis for future science fair or research competition submissions beyond IARCO, and to reach out to a genetics or developmental biology lab to ask whether the guide RNA candidates identified here could be tested experimentally, since that would be the next step this proposal points to. Regardless of the outcome, I plan to write up the epigenomic-analysis pipeline itself as a short methods note, so the approach could be reused for other monogenic disease targets by other researchers.

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